

[Art of Problem Solving](#)[AoPS Online](#)[Beast Academy](#)[AoPS Academy](#)

Euclidean Geometry In Mathematical Olympiads

[Bookmark](#) [New Topic](#)



Reply

#1

IMO 2006 Q1

$$\angle PBA + \angle PCA = \angle PBC + \angle PCB.$$

Show that $AP \geq AI$ and that equality holds if and only if $P = I$.

#2

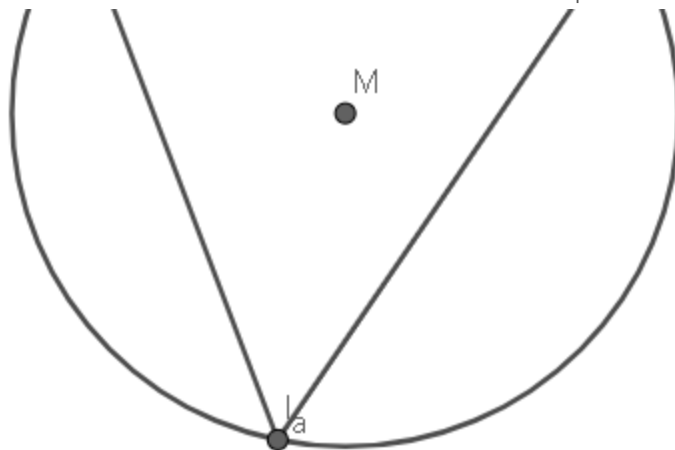
My solution

 Let

$$\angle PBA = \alpha, \angle PCA = \beta, \angle PBC = \gamma, \angle PCB = \delta.$$

$$\angle BCI_a = 90 - \frac{\beta + \delta}{2} \text{ and } \angle CBI_a = 90 - \frac{\alpha + \gamma}{2}.$$
$$AP + PM \geq AM = AI + IM \text{ (since APM could be degenerate)}$$
$$\overline{AP \geq AI}$$

,As required.



*This post has been edited 3 times. Last edited by JengaBetterThanLego, Dec 16, 2018, 11:20 AM
Reason: Typo.*

Kagebaka

3001 posts

Oct 16, 2018, 9:17 AM • 2 👍

Solution

#3